

Homework 1

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1.1

$(x + jy) = \text{cartesian form}$

$$\textcircled{1} \frac{1}{2} e^{j\pi} \Rightarrow \frac{1}{2} (\cos \pi + j \sin \pi) = \frac{1}{2} (1 + j0) = \frac{1}{2}$$

$$= -\frac{1}{2} + j0 = -\frac{1}{2}$$

$$\textcircled{2} \frac{1}{2} e^{-j\pi} \Rightarrow \frac{1}{2} (\cos(-\pi) + j \sin(-\pi)) = \frac{1}{2} (-1 + j0) = -\frac{1}{2}$$

$$(-1 + j0) = -\frac{1}{2} + j0 = -\frac{1}{2}$$

$$\textcircled{3} e^{j\frac{\pi}{2}} \Rightarrow \cos \frac{\pi}{2} + j \sin \left(\frac{\pi}{2} \right) = 0 + j \cdot 1 = j$$

$$\textcircled{4} e^{-j\frac{\pi}{2}} \Rightarrow \cos \left(-\frac{\pi}{2} \right) + j \sin \left(-\frac{\pi}{2} \right) = 0 + (-1)j = -j$$

$$\textcircled{5} e^{j\frac{5\pi}{2}} \Rightarrow \cos \frac{5\pi}{2} + j \sin \left(\frac{5\pi}{2} \right) = 0 + j \cdot 1 = j$$

$$\textcircled{6} \sqrt{2} e^{j\frac{\pi}{4}} \Rightarrow \sqrt{2} \left(\cos \left(\frac{\pi}{4} \right) + j \sin \left(\frac{\pi}{4} \right) \right) = \sqrt{2} \left(\frac{\sqrt{2}}{2} + j \frac{\sqrt{2}}{2} \right) = 1 + j$$

$$\textcircled{7} \sqrt{2} e^{j\frac{3\pi}{4}} \Rightarrow \sqrt{2} \left(\cos \left(\frac{3\pi}{4} \right) + j \sin \left(\frac{3\pi}{4} \right) \right) = \sqrt{2} \left(-\frac{\sqrt{2}}{2} + j \frac{\sqrt{2}}{2} \right) = -1 + j$$

$$\textcircled{8} \sqrt{2} e^{-j\frac{3\pi}{4}} \Rightarrow \sqrt{2} \left(\cos \left(-\frac{3\pi}{4} \right) + j \sin \left(-\frac{3\pi}{4} \right) \right) = \sqrt{2} \left(\frac{\sqrt{2}}{2} - j \frac{\sqrt{2}}{2} \right) = 1 - j$$

$$\textcircled{9} \sqrt{2} e^{-j\frac{\pi}{4}} \Rightarrow \sqrt{2} \left(\cos \left(-\frac{\pi}{4} \right) + j \sin \left(-\frac{\pi}{4} \right) \right) = \sqrt{2} \left(\frac{\sqrt{2}}{2} - j \frac{\sqrt{2}}{2} \right) = 1 - j$$

1.2

$(z e^{j\theta} \quad \pi < \theta \leq \pi) \Rightarrow \text{polar form}$

$$z = \sqrt{x^2 + y^2} \quad \theta = \tan^{-1} \left(\frac{y}{x} \right)$$

$$\textcircled{1} 5 \Rightarrow 5 + j0, \quad z = \sqrt{5^2 + 0^2} = 5, \quad \theta = \tan^{-1} \left(\frac{0}{5} \right) = 0$$

polar form = $5e^{j0}$

$$\textcircled{2} -2 \Rightarrow -2 + j0, \quad z = \sqrt{4 + 0^2} = 2, \quad \theta = \pi$$

on negative real axis

$$\textcircled{3} -3j \Rightarrow 0 - j3, \quad z = \sqrt{0^2 + 9} = 3, \quad \theta = -\frac{\pi}{2}$$

on negative imaginary axis

$$\textcircled{4} \frac{1}{2} - j\frac{\sqrt{3}}{2} \Rightarrow z = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1, \quad \theta = \tan^{-1} \left(\frac{-\frac{\sqrt{3}}{2}}{\frac{1}{2}} \right) = -\frac{\pi}{3}$$

$\theta = \tan^{-1}(-\sqrt{3}) \quad \theta = -\frac{\pi}{3}$

$$③ 1+j \Rightarrow z = \sqrt{1^2 + 1^2} = \sqrt{2} \quad \theta = \tan^{-1}(1) \\ \text{Polar form: } \sqrt{2} e^{j\frac{\pi}{4}} \quad \theta = \frac{\pi}{4}$$

$$⑥ (1-j)^2 \Rightarrow 1 - 2j + j^2 \Rightarrow -2j$$

$$j^2 = -1 \quad r = \sqrt{0^2 + (-2)^2} = 2 \\ 2 e^{-j\frac{\pi}{2}} \quad \text{on the negative imaginary axis}$$

$$④ j(1-j) \Rightarrow j - j^2 \Rightarrow j + 1$$

$$j^2 = -1 \quad z = \sqrt{1^2 + 1^2} = \sqrt{2} \quad \theta = \tan^{-1}(1) \\ \sqrt{2} e^{j\frac{\pi}{4}} \quad \theta = \frac{\pi}{4}$$

$$② \frac{(1+j)}{(1-j)} \Rightarrow \frac{(1+j)(1+j)}{(1-j)(1+j)} = \frac{1 + 2j + j^2}{1 - j^2} = \frac{2j}{2} = j$$

$$z = \sqrt{0^2 + 1^2} = 1 \quad \theta = \frac{\pi}{2} \quad \text{on positive imaginary axis} \\ e^{j\frac{\pi}{2}}$$

$$⑤ \frac{(\sqrt{2} + j\sqrt{2})}{(1 + j\sqrt{3})} \Rightarrow \frac{(\sqrt{2} + j\sqrt{2})(1 - j\sqrt{3})}{1 - 3j^2} \Rightarrow \\ \Rightarrow \frac{\sqrt{2} - j\sqrt{6} + j\sqrt{2} - j^2\sqrt{6}}{1 - 3(-1)} = \frac{\sqrt{2} - j\sqrt{6} + j\sqrt{2} + \sqrt{6}}{4} =$$

$$= \frac{\sqrt{2} - j\sqrt{6} + j\sqrt{2} + \sqrt{6}}{4} \Rightarrow \frac{\sqrt{2} + \sqrt{6}}{4} +$$

$$+ j \frac{\sqrt{2} + \sqrt{6}}{4}$$

$$z = \sqrt{\left(\frac{\sqrt{2} + \sqrt{6}}{4}\right)^2 + \left(\frac{\sqrt{2} + \sqrt{6}}{4}\right)^2} = \sqrt{\frac{16}{16}} = 1$$

$$\tan^{-1}\left(\frac{\frac{\sqrt{2} - \sqrt{6}}{4}}{\frac{\sqrt{2} + \sqrt{6}}{4}}\right) = \tan^{-1}\left(\frac{\sqrt{2} - \sqrt{6}}{\sqrt{2} + \sqrt{6}}\right)$$

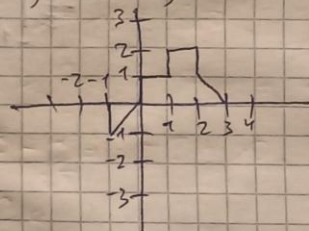
$$\theta = \frac{\pi}{12}$$

$$e^{-j\frac{\pi}{12}}$$

$$(1.27)$$

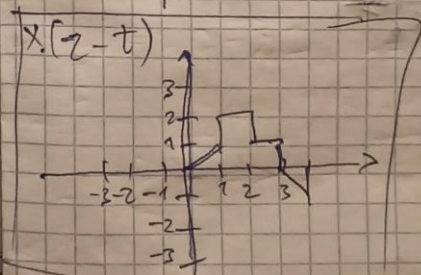
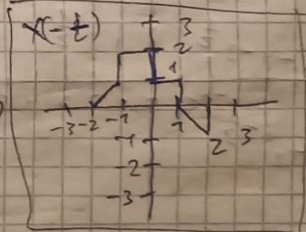
$$a) x(t-1)$$

$t=1$ - right shift

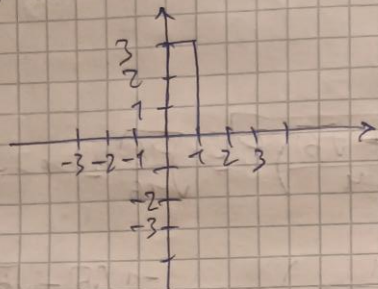


$$⑧ x(2-t) \Rightarrow \text{reverse}$$

$$2-t=0 \quad t=2 \quad \text{2-shift}$$

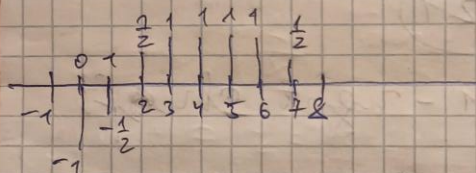


e) $(x(t) + x(-t)) \cdot u(t)$

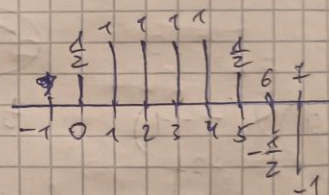


1.22

a) $x(n-u)$



b) $x(3-n)$



c) $x(n) \cdot u(3-n) \Rightarrow x(n)$

